Life History Strategies

Pacific salmon

Red kangaroo

Mayfly

Bamboo

Life History

Life history: species lifetime pattern of growth, development and reproduction.

A measure of organism's reproductive success is fitness

Trade-off between growth and reproduction:

--- mode of reproduction, age at rep., allocation to rep. number and size of eggs, young or seeds, parental care.

Life History Patterns

- Modes of reproduction: sexual or asexual
- Forms of Sexual reproduction
- Mating systems
- Mate selection
- Females may acquire mates based on resources
- Organisms budget time and energy to reproduction

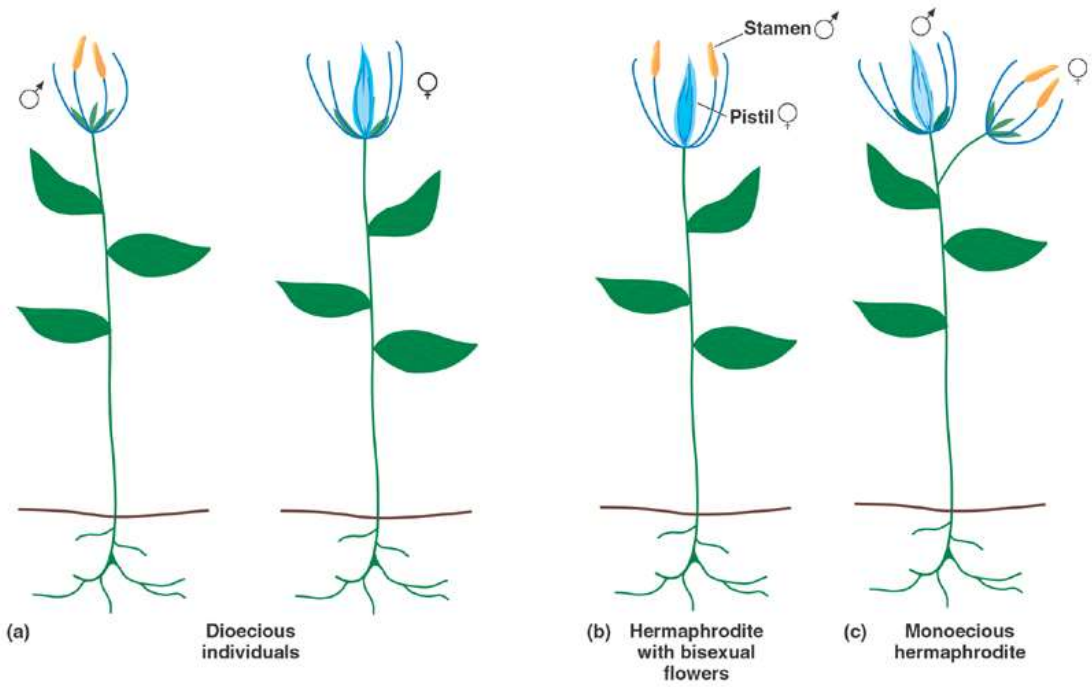
- Timing of reproduction
- Parental investment
- Fecundity depends on age and size
- Food supply affects the production of young

Sexual or Asexual Reproduction

- Asexual reproduction (produce offspring without involving of egg and sperm)
  - New individuals are the same as the parent
  - Many plants (underground stem) such as strawberry;
  - some animals (hydra, some aphids, parthenogenesis)
  - If fitness is high, matches organism to environment
  - If fitness is low, possible extinction (less variation)

- Sexual Reproduction
  - More common form.
  - Can produce new gene combinations able to cope with a changing environment.
  - Greater energy commitment
    - Specific organelles
    - Production of gametes, courtship activities, and mating are energetically expensive.
    - Feeding offspring
    - The expense of reproduction is not shared equally by both sexes

Types of sexual reproduction



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- Dioecious
- Hermaphroditic
- Monoecious

Floral structure

Plants

Animals

- Most familiar form involves male and female individuals

- Hermaphroditic
  - Simultaneous hermaphroditic
    - Earthworms
    - Outbreeding, but maximizes offspring (twice)

  - Sequential hermaphroditic

    - Mollusks, echinoderms Sometimes vertebrates (fish)
    - Allows all individual to participate in both sides of sexual cycle

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Parrotfish

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Largest female becomes male when the male fish is missing

Mating Systems

- Different mating strategies have different advantages and disadvantages
  - Monogamy (one to one, form of a lasting pair bond between one male and one female)
    - Most prevalent among birds, rare among mammals
    - Seasonal or permanent
      - Allows sharing of cost of raising offspring
      - Increases survival chances of offspring
      - Cheating does occur and has specific advantages to fitness
  - Polygamy (one to two or more, a pair bond exists between individual and each mate)
    - More than one mate of one sex for a single individual of the other sex (polygyny and polyandry)
      - Free individual to compete for resources and protect territory
      - Better food etc for mates
      - Some protection of offspring from competition
  - Promiscuity (one to one or many and no pair bond formed)
    - Greatest number of offspring
    - Large amount of competition
    - Female only responsible for offspring in terms of resources
      - Poorer survival chance for offspring

Sexual Selection

- **For Monogamy, Polygamy and Promiscuity**
  - **All involve the selection of a mate and therefore sexual selection**
  - **Selection for secondary sexual characteristics**

  - Examples
    - **Peacock versus Peahen**
      - **Large tail feathers, more mating**
      - **Smaller tail feathers, less mating**
    - **Deer**
      - **Characters that aid competition such as horns**
    - **Humans**
      - **Faster sports car such as a Ferrari**

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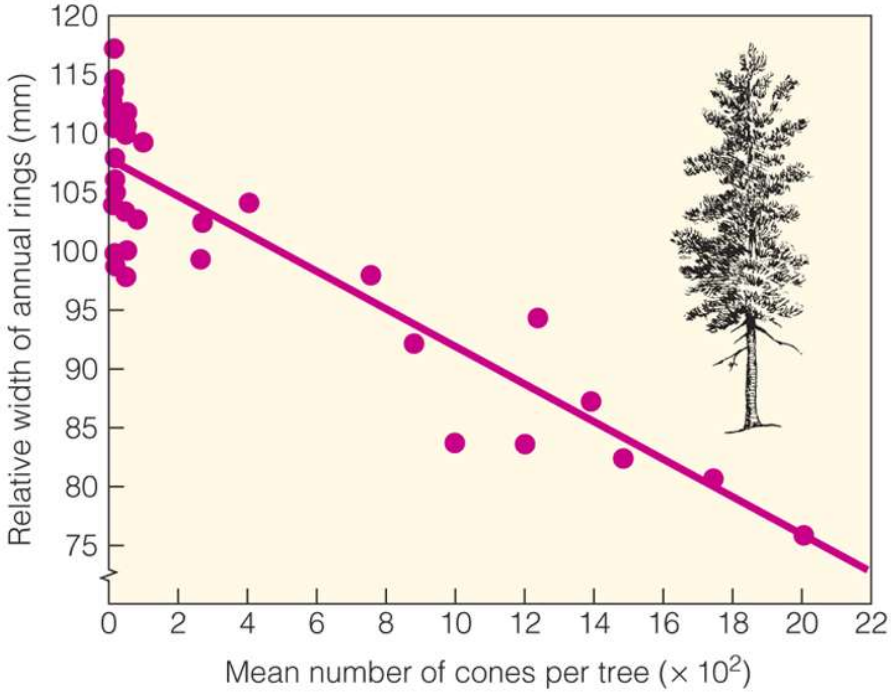
Organisms budget time and energy to reproduction

- Reproductive effort: Time and energy allocated to reproduction

- Trade-off between growth, maintenance and reproduction.

Percentage of annual production to reproduction:

Perennials: 15-20%
Wild annuals: 15-30%
Crops: 25-30%
Corn and barley: 35-40%
Lizard: 7-9%
Salamander: 48%



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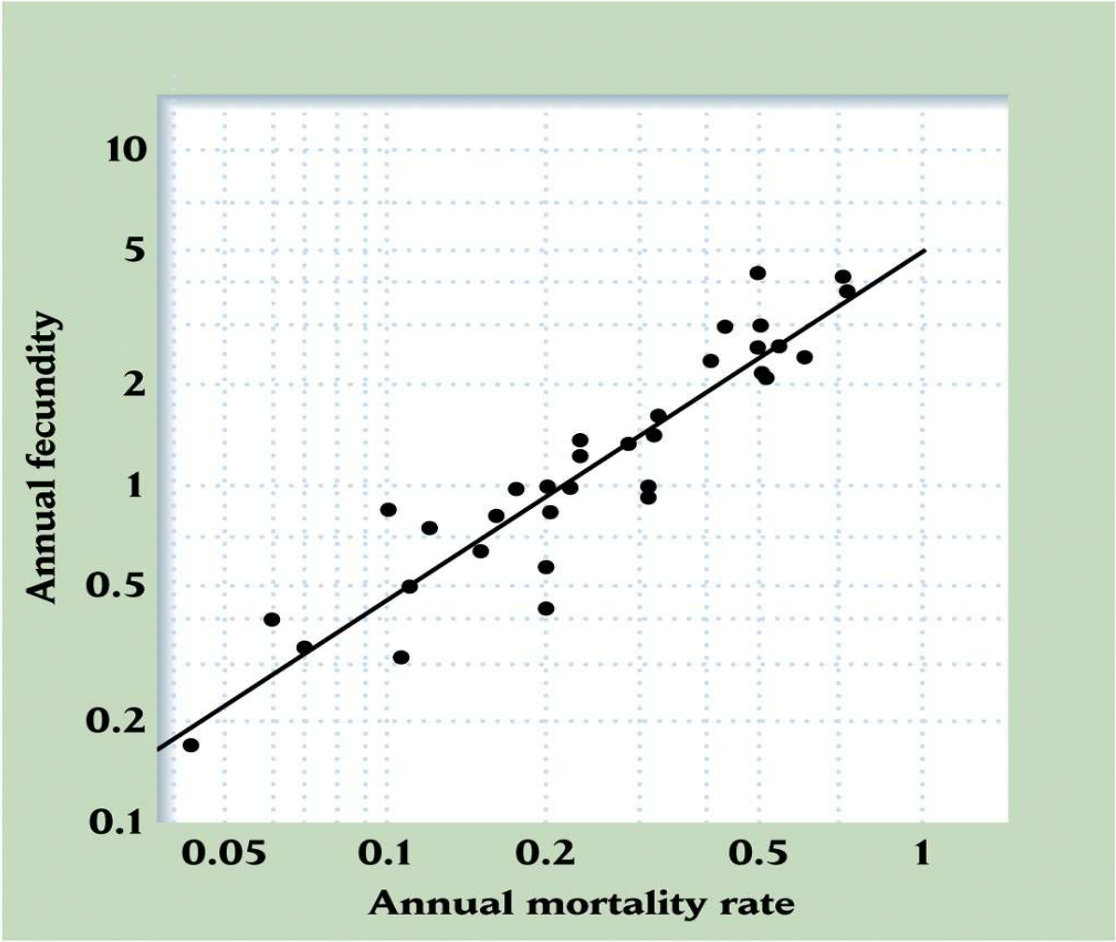
Bird species illustrate the tradeoff between “fast” organisms (high fecundity, high mortality) versus “slow” ones (long life, low annual fecundity)



Species differ in timing of reproduction

- Semelparity
  - One reproductive effort with all resources, then death
  - Most insects and other invertebrates, some fish (salmon) and many plants (bamboo, ragweed)
  - Some are small, short lived, grown in disturbed habitats;
  - Environmental effect can be disastrous

- Iteroparity
  - Produce fewer young at one time and repeat reproduction throughout their lifetime
  - Multiple cycles of reproduction means the organism must balance growth, maintenance, escaping predators, defending territory, etc against reproduction
  - Most vertebrates, perennial herbaceous plants, shrubs, and trees.

Sockeye salmon, for example, swim as far as 6,000 km from Pacific Ocean from feeding grounds to spawning streams, lay thousands of eggs, then die from the exertion.

Parental investment: depends on the number and size of offspring

- Given certain resource allocated to rep., one can produce many small young or few large ones.

- Produce large number of offspring, less or no parental care (fish-eggs, plants-seeds)
- Produce helpless offspring (produce young, spend less energy in incubation, but require considerable parental care)

  Altricial
  - Mice
    - Longer period suckling
  - American Robin
    - Other bird feeds
- Produce more mature offspring (longer gestation, born in advanced stage of development)

  Precocial
  - Chicken, cow, deer, turkey

  Humans ?
    - Family care (Grandmothers, Grandfathers, Aunts, Uncles, Brothers and Sisters)

African elephants produce one offspring at a time, once every few years over a long lifetime, and protect each offspring intensively (much like humans)

Many plants and some insects reproduce once (annually), producing vast numbers of seeds/eggs that are poorly protected, if at all

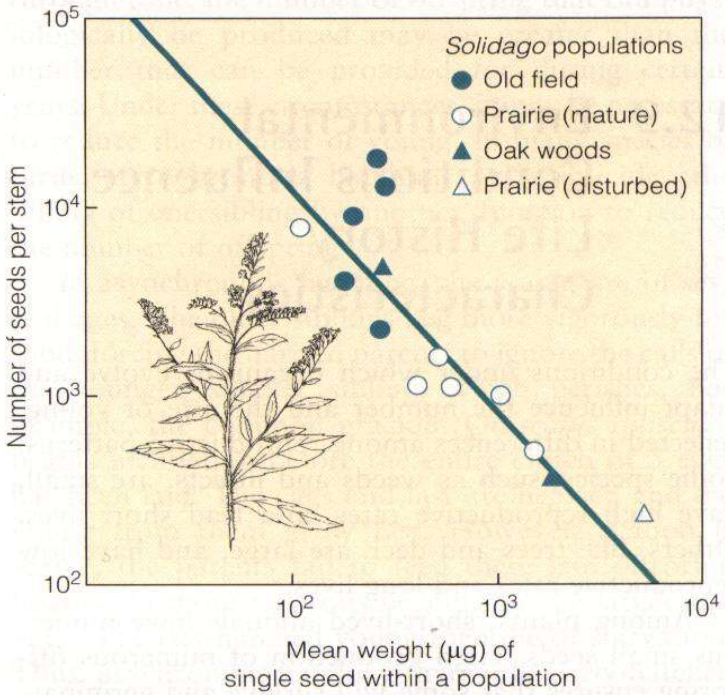


FIGURE 12.8 Inverse relationship between mean weight of individual seeds and the number of seeds produced per stem for populations of goldenrod *(Solidago)* in a variety of habitats. (Werner and Platt 1976.)

Large Number

Less resources per individual

More chances of success

Fecundity depends on age and size

- For many species, number of offspring produced varies with the age and size of parent.

- But some species do not have a characteristic adult size and can continue to grow through their adults lives (***indeterminate growth***)
  - Many plants and ectothermic (cold-blooded) animals (fish, reptiles, amphibians, and invertebrates)

Plants

Perennial: delay flowering until they have attained a sufficiently large size

Annual: small plants produce less seeds

Animals:

Ectothermic (cold-blooded) animals

Production of offspring in fish increases with size, which increases with age

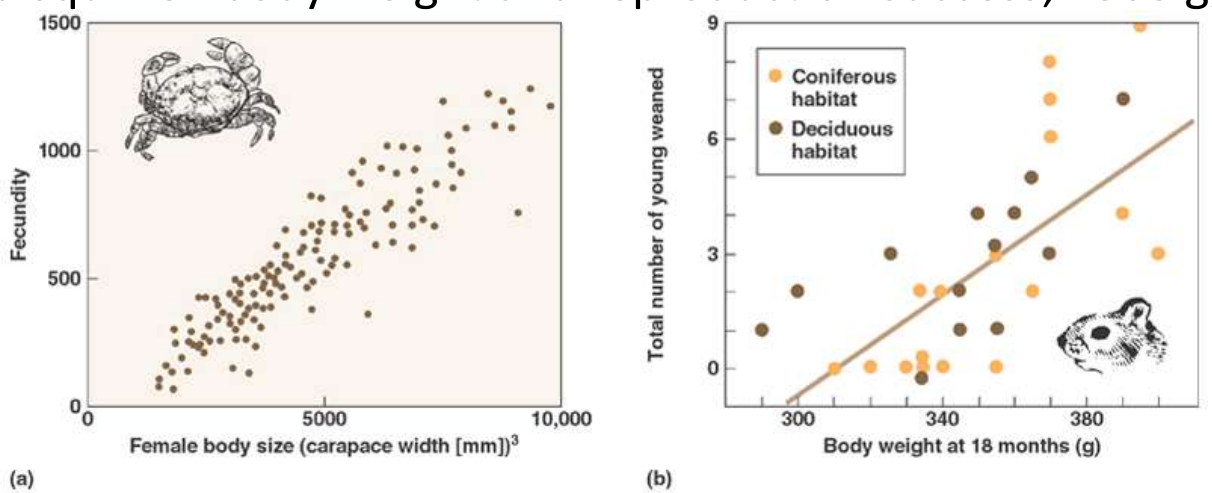
Gizzard shad: 2-yr, 59,000 eggs

3-yr, 379,000 eggs

Endothermic (warm-blooded):

similar patterns exist for some animals

European red squirrel: body weight and reproduction success; <300 g, do not reproduce.



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Environmental conditions influence the evolution of life history characteristics

- Idea was conceived by Robert MacArthur and Edward O. Wilson:
- "r- vs. K-selected strategists"

(a)

(b)

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Spotted and redback salamanders

Life history strategies

Different organisms have evolved different sets of traits related to population dynamics

Traits that affect an organism's schedule of reproduction and survival make up its <u>life history</u>

Energy is limited.

- Allocation to one function (such as growth) will reduce the amount of energy an organism can devote to another function (such as reproduction).

- An organisms life history strategy involves trade-offs.

Life history strategies

Different organisms have evolved different sets of traits related to population dynamics

Traits that affect an organism's schedule of reproduction and survival make up its <u>life history</u>

Annual plants

- Reproduction occurs once **(semelparity)**
- It occurs after one season
- Death occurs after reproduction
- Many small seeds are produced
- Favored in unstable environments

Life history strategies

Different organisms have evolved different sets of traits related to population dynamics

Traits that affect an organism's schedule of reproduction and survival make up its <u>life history</u>

Perennial plants

- Reproduction occurs more than once **(iteroparity)**
- It occurs after many seasons
- Larger seeds are produced (few well-provisioned seeds have higher survival rates)
- Favored in stable environments

K and r selection (MacArthur and Wilson 1967)

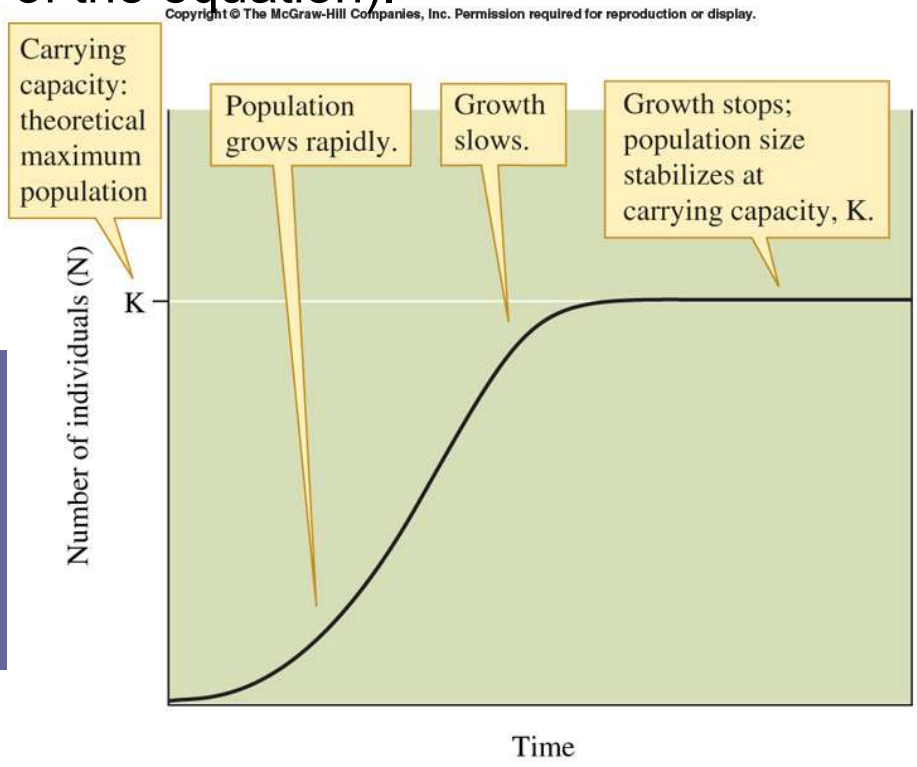
Based on two opposing selective forces **r- selection** and **K-selection**, after the two terms in the logistic equation (however, one should not take these terms too literally, as the concepts are independent of the equation).

r-selected species

- r refers to the per capita rate of increase
- Selection favoring rapid growth
- Should be favored in new or disturbed environments
- Less competition

K-selected species

- K refers to carrying capacity
- More prominent in species that are typically at their carrying capacity
- Favors more efficient use of resources
- Live with competition



Life history strategies

K and r selection (MacArthur and Wilson 1967)

Characteristics of r- v. K-selected species		
	r-selection	K-selection
Population Growth Rate		
Competitive Ability		
Pace of Development		
Body Size		
Type of Reproduction		
Environment		
Parental Care		
Survivorship		
Number of Offspring		

Life history strategies

K and r selection (MacArthur and Wilson 1967)

Characteristics of r- v. K-selected species		
	r-selection	K-selection
Population Growth Rate	high	low
Competitive Ability	low	high
Pace of Development	fast	slow
Body Size	small	large
Type of Reproduction	semelparity	iteroparity
Environment	disturbed	stable
Parental Care	little	A lot
Survivorship	Type III	Type I
Number of Offspring	many	few

Life History: refers to any aspect of the developmental pattern and mode of reproduction of an organism.